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PHNOM PENH INTERNATIONAL UNIVERSITY

Machine Learning

SMS Spam Detection

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Presentation Agenda

1

Problem Statement

Real-world SMS threats & ML vs rule-based filters.

2

Dataset Analysis

UCI SMS Spam collection breakdown & class distribution.

3

Preprocessing Pipeline

Cleaning, Stemming, & TF-IDF feature extraction.

4

Multinomial Naïve Bayes

Bayesian probabilistic model & Laplace hyper-parameters.

5

Logistic Regression

Linear decision boundary & L2 regularization tuning.

6

Results & Demo

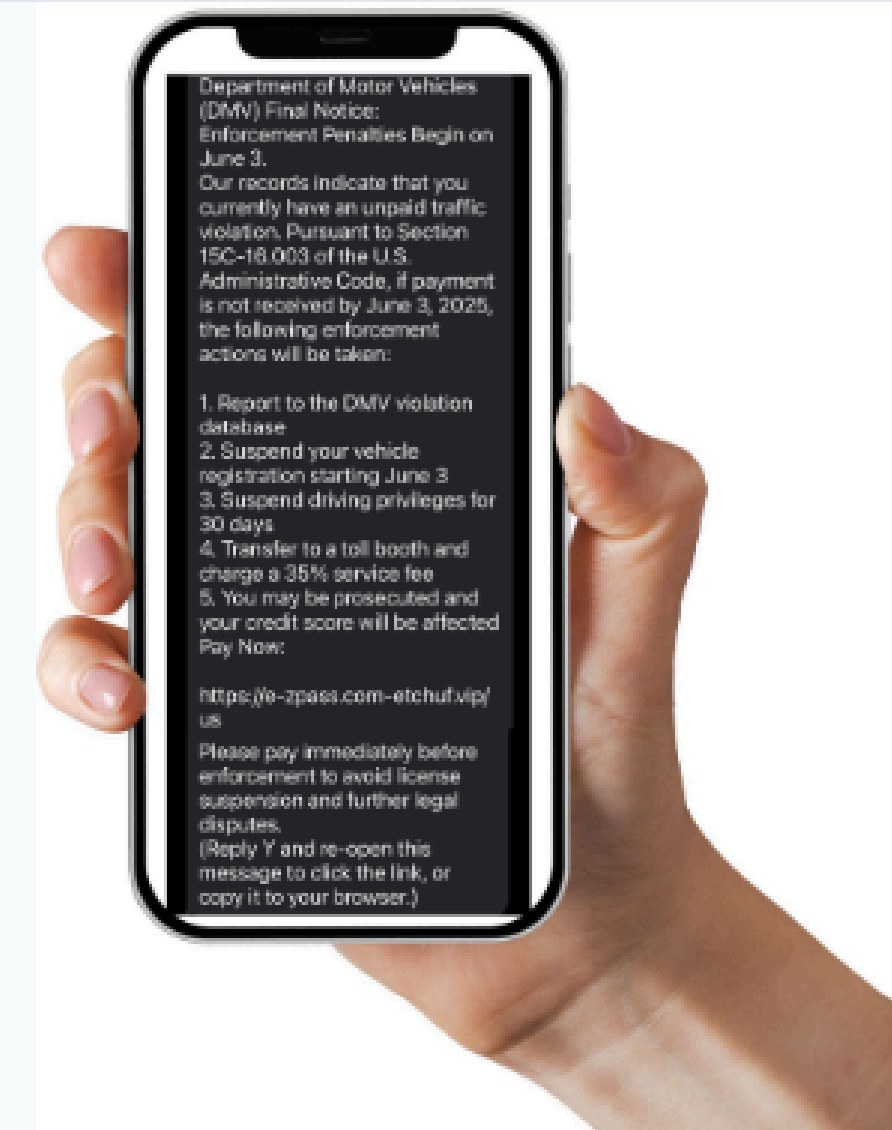
Performance comparison & interactive Gradio deployment.

Problem Statement & Threats

🛡️ The Real-World Threat

SMS spam poses severe cybersecurity risks including financial scams, identity theft, malware links, and deceptive phishing targeting mobile device users globally.

- ✓ **Rule-Based Limits:** Manual static keyword rules fail because spammers constantly adapt with typos, obfuscation, and evolving slang.
- ✓ **Machine Learning Advantage:** Adaptive probabilistic models learn underlying statistical patterns to detect novel, unseen spam variations.
- ✓ **Class Imbalance:** Binary task with heavy class skew (~86.6% HAM vs ~13.4% SPAM) requiring careful metric selection beyond accuracy.



Dataset: UCI SMS Spam Collection

5,574
TOTAL MESSAGES

4,827
LEGITIMATE (HAM) — 86.6%

747
SPAM MESSAGES — 13.4%

HAM SAMPLE

"Hey, are you coming to the party tonight? Let me know when you leave."

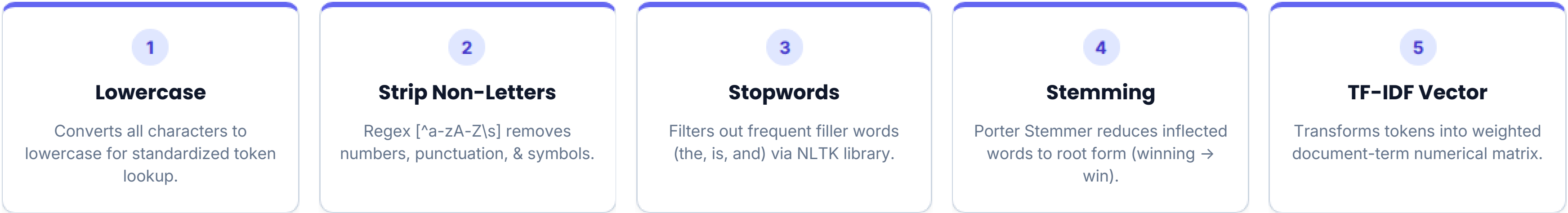
SPAM SAMPLE

"URGENT! Your bank account has been compromised. Send your password immediately to verify."



Text Preprocessing Pipeline

Standardized natural language processing pipeline converts noisy raw text into clean, numerical feature vectors prior to classifier training.



Stratified Train-Test Split

80% Training / 20% Testing split maintaining identical class ratios across splits. Replicated exactly in web inferencing.

EXACT PIPELINE REPLICATION

Algorithm — Naive Bayes

Naive Bayes is a machine learning classification algorithm based on probability. It comes from something called Bayes' Theorem, which helps us calculate the probability of something based on prior knowledge.

The diagram illustrates Bayes' Theorem with the following components and labels:

- Likelihood of the Evidence given that the Hypothesis is True**: Labeled in orange, with an arrow pointing to $P(E|H)$ in the numerator.
- Prior Probability of the Hypothesis**: Labeled in red, with an arrow pointing to $P(H)$ in the numerator.
- Posterior Probability of the Hypothesis given that the Evidence is True**: Labeled in blue, with an arrow pointing to $P(H|E)$ on the left side of the equation.
- Prior Probability that the evidence is True**: Labeled in green, with an arrow pointing to $P(E)$ in the denominator.

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$

Naive Bayes works by:

1. Looking at the words in a message
2. Calculating the probability of each word appearing in spam or non-spam messages
3. Combining these probabilities
4. Predicting whether the message is Spam or Not Spam

Algorithm — Naive Bayes

HYPERPARAMETER OPTIMIZATION

GridSearchCV Configuration

5-Fold Cross Validation evaluated over candidate smoothing parameter grid:

```
param_grid = {'alpha': [0.01, 0.1, 0.5, 1.0, 5.0]}
```

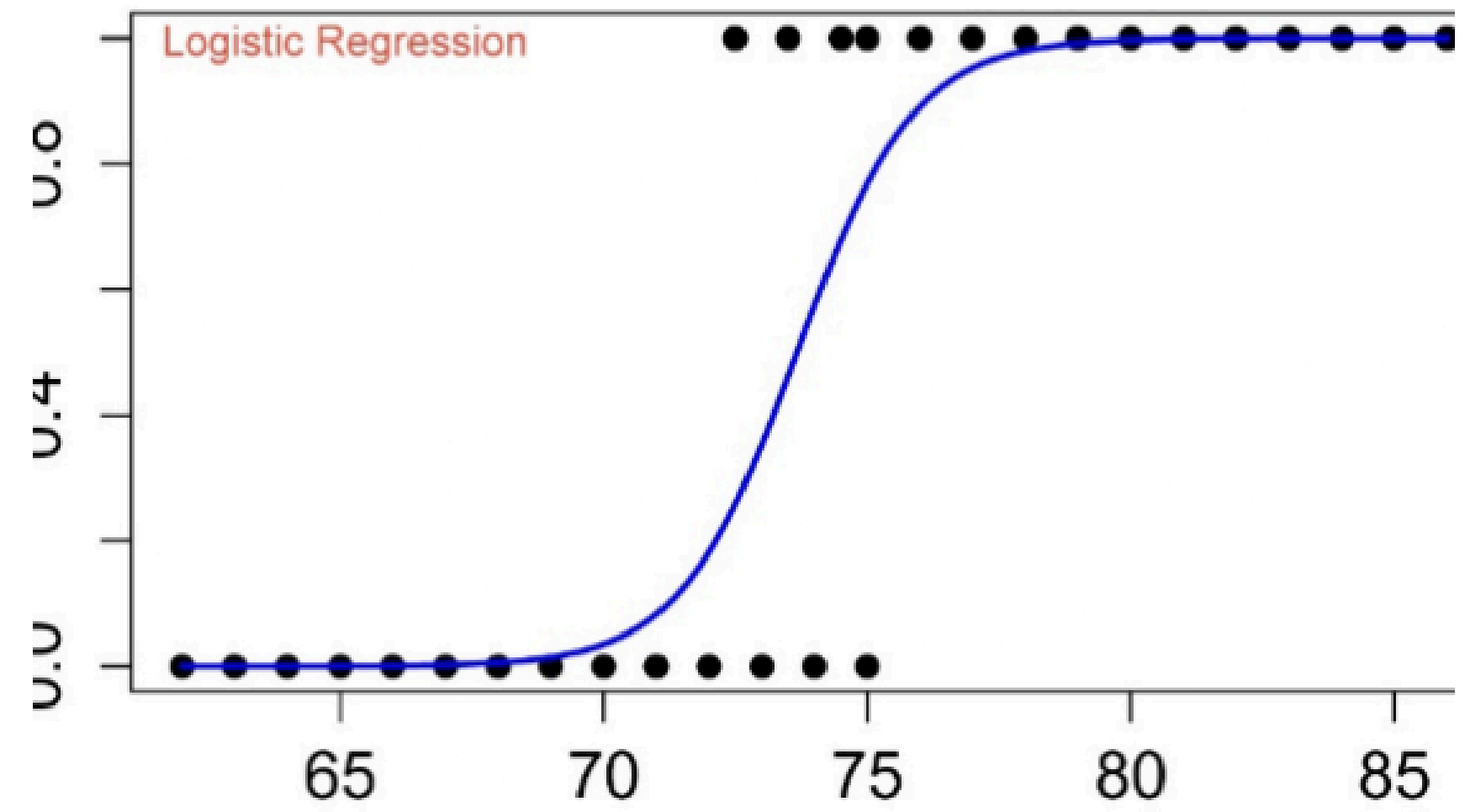
We use GridSearchCV to tune hyperparameters like alpha, so the model gives better spam detection accuracy

Metric	Value
Accuracy	0.9758
Precision	0.9773
Recall	0.8658
F1-Score	0.9032
ROC-AUC	0.9821

Algorithm — Logistic Regression

Logistic Regression models posterior class probability using a linear combination of TF-IDF feature weights squashed by the Sigmoid function:

$$P(y = 1 \mid x) = \sigma(w \cdot x + b) = \frac{1}{1 + e^{-w \cdot x + b}}$$



- Direct Probability Modeling: Outputs calibrated confidence scores between 0.0 and 1.0.
- No Independence Assumption: Seamlessly handles correlated feature pairs (e.g. "bank" & "account").
- L2 Penalty Support: Prevents overfitting on high-dimensional sparse text data.

Algorithm — Logistic Regression

Metric	Value
Accuracy	0.9785
Precision	0.9635
Recall	0.8859
F1-Score	0.9167
ROC-AUC	0.9838

Model Comparison & Results

Performance Metrics Benchmark

Metric	Naive Bayes	Logistic Reg.	Winner
Accuracy	0.9758	0.9785	Logistic Reg.
F1-Score	0.9032	0.9167	Logistic Reg.
ROC-AUC	0.9821	0.9838	Logistic Reg.

Structural Model Differences

Feature	Naive Bayes	Logistic Reg.
Core Rule	Bayes Theorem	Linear + Sigmoid
Feature Indep.	Assumed (Naive)	Not Assumed
Training Speed	Fast (1-pass)	Iterative GD

Interactive Gradio Demo

Gradio Demo

Thank you!